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Nanotribology of Ionic liquids as Lubricant Additives for Alumina Surfaces

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Abstract

Common lubricants, optimized for steel, perform poorly for alumina surfaces, making ionic liquids (ILs) attractive potential alternatives, either in neat form or as oil additives. Here, atomic force microscopy (AFM) has been used to study the lubricity of two oil – miscible ILs between a silicon AFM tip and an alumina surface. Trihexyltetradecylphosphonium bis(2,4,4-trimethylpentyl)phosphinate ($P_{6,6,6,14} (iC_8)_2PO_2$) and trihexyltetradecylphosphonium bis(2-ethylhexyl) phosphate ($P_{6,6,6,14}$ DEHP) were mixed with hexadecane at concentrations from 0 to 100 mol% IL. Both ILs are effective lubricants for alumina, reducing lateral forces to approximately a third of the forces measured in hexadecane. When used as an additive in hexadecane, increasing the IL concentration generally decreased the friction, as the IL adsorbs on alumina and forms a strong boundary layer. $P_{6,6,6,14}$ DEHP mixtures reduce friction more effectively than $P_{6,6,6,14} (iC_8)_2PO_2$. For both ILs, a 2 mol% mixture of IL and hexadecane reduced friction most effectively, even more than the neat IL.

Introduction

Lubricants reduce friction, adhesion and wear between sliding surfaces. Under low loads, viscous hydrodynamic forces separate the surfaces and prevent physical contact.¹ At high loads, the bulk of the lubricant is squeezed out, and only a single boundary layer of lubricant molecules remains between the sliding surfaces.² Common hydrocarbon-based oils adsorb weakly to surfaces, so oil-miscible surfactants and polymers are added which adsorb to the surface and remain in place even under high loads.¹⁻² The effectiveness of surfactant and polymer additives depends on the affinity it has for the surface via Coulombic, van der Waals, H-bonding, and solvophobic interactions. Unfortunately, many common additives which are effective on steel surfaces are not effective on light-weight surfaces such as aluminum and aluminum alloys.³⁻⁴ Thus, there is a push towards developing additives to match these surfaces.

Ionic liquids (ILs) are pure salts that have melting points below 100 °C.⁵ ILs are high performing lubricants in neat (undiluted) form for a variety of surfaces, including silica, steel, aluminum, titanium at both the macro-⁶⁻¹² and nano-¹³⁻¹⁷ scales. Like conventional high-end lubricants, such as perfluoropolyethers, ILs have high thermal stability and low vapor pressure, enabling them to operate in extreme conditions, e.g. high temperature and low pressure. Additionally, IL ions interact strongly with surfaces and each other via Coulombic, van der Waals, H-bonding and solvophobic interactions¹⁸⁻¹⁹; this means they resist being ‘squeezed-out’, meaning a lubricating film remains in place up to higher loads than for a comparable conventional lubricant.^{14, 16-17, 20-25}

As the strength of IL boundary layer is related to chemical composition and structure (i.e. alkyl chain length and charge localization), control of IL lubricity can be achieved by tuning the composition and orientation of ions in the boundary layer by changing IL molecular structure and/or surface properties.

ILs are relatively expensive compared with current commercially available lubricants. However, the recent development of oil miscible ILs means they can be diluted in cheaper base oils and used as additives.²⁶⁻³⁴ IL/oil mixtures investigated at the macroscale outperform conventional additives at comparable concentrations,²⁶⁻³³ but the lubrication mechanisms of using ILs as additives remain only partially understood.²⁷ Nanotribology studies are less affected by surface asperities and roughness than macroscale measurements, and thus are able to reveal lubrication mechanisms.²¹ Recent macro- and nanotribology studies have demonstrated the efficiency of two oil miscible ILs, trihexyl(tetradecyl)phosphonium bis(2,4,4-trimethylpentyl)phosphinate ($P_{6,6,6,14} ({}^iC_8)_2PO_2$) and trihexyltetradecylphosphonium bis(2-ethylhexyl) phosphate ($P_{6,6,6,14}$ DEHP) as neat lubricants and as lubricant base oil additives.^{11, 27} Nanotribology experiments of $P_{6,6,6,14} ({}^iC_8)_2PO_2$ /hexadecane mixtures on silica and titanium surfaces reveal that a minimum concentration of IL ($\sim 1 - 2$ mol%) is required to form a robust boundary layer.

Due to its high strength to weight ratio, aluminum and its alloys have been widely applied in capacitors, lightweight mechanical components, and optical/analytical devices.³⁵⁻³⁷ Furthermore, aluminum forms a passivating oxide layer in air that protects the bulk aluminum from further corrosion, unlike iron, which corrodes destructively.³⁵⁻³⁷ One of the drawbacks of aluminum is that common lubricants designed for steel surfaces cannot lubricate alumina surfaces effectively, leading to high friction and wear.^{9, 27, 38}

The alumina surface is net positively charged in water³⁹ at neutral pH, and based on past experience,⁴⁰ a net positive charge is expected to persist in IL/oil mixtures. This contrasts with previously investigated net negatively charged silica and titania surfaces,⁴⁰⁻⁴¹ and thus the adsorbed boundary layer on alumina is expected to be anion rich rather than cation rich like for silica and titania. Macrotribology investigations for ILs as neat lubricants^{8, 11-12} and IL/oil

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3 mixtures as lubricant additives^{27,34} on aluminum-steel interfaces have shown the effectiveness of
4 these ILs. However, the lubrication mechanisms are currently poorly understood. No
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6 fundamental nanotribology studies have been conducted on alumina surfaces, especially in the
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8 boundary layer lubrication regime, where the bulk lubricant film has been squeezed out and only
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10 remains a surface adsorbed boundary layer. The effects of ion structure and concentration on the
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12 adsorbed boundary layer and lubricity are also unclear.
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17 In this work, we investigate the nanotribology of two oil miscible ILs, $P_{6,6,6,14}^i(C_8)_2PO_2$ and
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19 $P_{6,6,6,14}$ DEHP, as lubricant additives for alumina surfaces in the boundary layer lubrication
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21 regime. The outcomes reveal the relationship between IL structure and boundary layer
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23 lubrication, and thus help to elucidate the lubrication mechanisms of IL/oil mixtures and aid the
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25 development of new, cost-effective lubricant additives for aluminum and aluminum alloy
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27 systems.
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31 Materials and Methods

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34 Flat alumina surfaces (10 mm × 10 mm × 0.5 mm, purity 99.6%) were purchased from MTI
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36 Corporation, CA. The morphology of the alumina surface was investigated using contact mode
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38 AFM with a scan size of 1 μm × 1 μm, cf. Figure S1; the surface roughness is 4±1 nm.
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40 Hexadecane is used as a model apolar lubricant base oil due to its simple structure, well-known
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42 properties, and commercial availability in high purity. Hexadecane (ReagentPlus®, 99%) was
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44 purchased from Sigma-Aldrich. Both ILs, $P_{6,6,6,14}$ DEHP (purity > 98%) and $P_{6,6,6,14}^i(C_8)_2PO_2$
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46 (purity > 95%), provided by Iolitec, are chemically stable in air and fully miscible in non-polar
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48 base oils. The zero shear viscosities of the IL/hexadecane mixtures at 20 °C were measured by an
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50 AR-G2 rheometer (TA instruments), and are provided in Table S1 and Table S2.
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All nanotribological measurements were determined using a Bruker Nanoscope MultiMode 8 AFM with an EV scanner in contact mode. AFM cantilevers (spring constant = 2.0 ± 0.2 N/m by the thermal tune method, tip radius ~ 8 nm) from the same batch (model NSC36 with Al backside coating, Mikromasch, Tallinn, Estonia) were used over the course of the investigation. The cantilevers were cleaned with ethanol and Milli-Q water (resistivity $18.2 \text{ M}\Omega\cdot\text{cm}$ at 25°C) dried under nitrogen and irradiated with UV light for at least 15 min before use. The sample volume of the AFM liquid cell was 0.1 mL. 0.5 mL of the sample was flushed in the AFM liquid cell before equilibrating for 20 min.

Friction (lateral) forces were obtained by performing AFM scans with a scan angle of 90° (with respect to the long axis of the cantilever) and with the slow scan axis disabled.⁴² The scan size was 100 nm, and scan rate was 30 Hz. The lateral deflection signal (i.e., cantilever twist) was converted to lateral force using a customized function produced in Matlab R2015a which takes into account the torsional spring constant and the geometrical dimensions of the cantilever. These experiments were repeated more than three times; the resulting lateral force data showed the same trend with small variations. Examples of repeated results for 0.5 mol% $\text{P}_{6,6,6,14}$ ($^i\text{C}_8$) $_2\text{PO}_2$ /hexadecane and 0.5 mol% $\text{P}_{6,6,6,14}$ DEHP/hexadecane mixtures are shown in Figure S2.

Results and Discussion

The nanotribology of the IL, hexadecane and various IL/hexadecane concentrations was investigated with AFM. The lateral (friction) force as a function of normal load up to 300 nN is presented in Figure 1 and Figure 2. Loads in mechanical engine parts and disc brakes generally do not exceed 0.5 GPa, which corresponds to a load of ~ 70 nN in this study.^{41, 43} Loads up to 300 nN therefore probe IL lubrication for extreme conditions. Friction coefficients (μ) were calculated using a modified version of Amontons' Law, $F_L = F_L(0) + \mu F_N$, where $F_L(0)$ is the

lateral force at zero applied normal load, dependent on the adhesion, roughness and compressibility of both the AFM tip and surface.⁴⁴

Hexadecane

The lateral force data for the AFM tip sliding over the alumina surface in hexadecane is similar to previous studies on weakly absorbing liquids at various surfaces, including silica and titania^{41, 44-46}. At zero normal load there is a small lateral force (~10 nN) caused by adhesion between the silica tip and the alumina surface⁴⁷, consistent with the slightly attractive forces detected in the normal force-separation profiles, *cf.* Figure S3(a). Above 10 nN the lateral force increases at a constant rate, giving a friction coefficient of $\mu = 1.0 \pm 0.3$. An average data set for hexadecane on alumina is shown in Figure 1 and Figure 2; however, some variability (± 90 nN for lateral force measurements at a normal load of 300 N) was observed for hexadecane which was not observed for the IL/hexadecane mixtures. The variability is likely due to variation in surface roughness and/or humidity for different tests. As expected, hexadecane is a poor lubricant and shows high friction because it has weak interaction with alumina and thus adsorbs weakly to the alumina surface; therefore hexadecane is easily pushed through by the AFM tip even at low normal loads, resulting in direct contact of the AFM tip and the alumina surface.

$P_{6,6,6,14} ({}^iC_8)_2PO_2$

Results for lateral force vs normal load data for $P_{6,6,6,14} ({}^iC_8)_2PO_2$ on alumina reveal excellent lubrication, similar to previous studies on silica and titania.^{41, 44, 46} Unlike in hexadecane, at zero normal load negligible lateral force is measured in $P_{6,6,6,14} ({}^iC_8)_2PO_2$, in line with negligible attraction detected in the normal force-separation data provided in Figure S3(b). The lateral forces increase with normal load at a constant rate ($\mu = 0.25 \pm 0.01$) which is around three times lower than that of hexadecane. The much lower lateral force and friction coefficient for $P_{6,6,6,14}$

$(^i\text{C}_8)_2\text{PO}_2$ is consistent with the formation of an IL boundary layer formed by $\text{P}_{6,6,6,14} (^i\text{C}_8)_2\text{PO}_2$ ions adsorbed to the alumina surface.^{11, 27} However, at a normal load of ~ 230 nN, lateral forces of $\text{P}_{6,6,6,14} (^i\text{C}_8)_2\text{PO}_2$ begin to increase at a faster rate ($\mu = 0.55 \pm 0.04$). This suggests a change takes place in the boundary layer between the AFM tip and the alumina surface at high loads. The change is consistent with the rupture of the lubricating boundary layer by the AFM tip at high loads, as proposed previously for the same IL on titania surfaces.⁴¹ Interestingly, although the surface roughness of alumina is higher than that of silica and titania used in previous studies, the friction coefficient of $\text{P}_{6,6,6,14} (^i\text{C}_8)_2\text{PO}_2$ on alumina ($\mu = 0.25 \pm 0.01$) before the breakpoint is slightly lower than that on silica ($\mu = 0.34 \pm 0.03$) and titania ($\mu = 0.40 \pm 0.04$). Silica and titania are negatively charged surfaces, therefore the IL boundary layer on these surfaces is expected to be cation-rich. Alumina, in contrast, is net positively charged, and therefore the IL boundary layer is anion-rich. This indicates that the $(^i\text{C}_8)_2\text{PO}_2^-$ anions at positively charged alumina surfaces are more lubricating than the $\text{P}_{6,6,6,14}^+$ cations at negatively charged silica and titania surfaces.

Lateral forces for 0.1 mol% $\text{P}_{6,6,6,14} (^i\text{C}_8)_2\text{PO}_2$ in hexadecane ($\mu = 0.95 \pm 0.04$) are similar to hexadecane ($\mu = 1.0 \pm 0.3$) with reduced variability. An earlier study showed that the IL surface excess increases with the bulk IL concentration.⁴⁶ At 0.1 mol%, the $\text{P}_{6,6,6,14} (^i\text{C}_8)_2\text{PO}_2$ surface excess at the alumina interface is too low to form an effective boundary layer; the silica tip likely slides directly on the surface as it does in hexadecane, resulting in similar lateral force data to hexadecane. The improved reproducibility indicates the added IL assists to reduce the variation in surface roughness and adhesive forces.

When the $\text{P}_{6,6,6,14} (^i\text{C}_8)_2\text{PO}_2$ concentration is increased to 0.5 mol%, at low normal loads (< 130 nN), the lateral force increases slowly and is comparable to the neat $\text{P}_{6,6,6,14} (^i\text{C}_8)_2\text{PO}_2$. However,

when the normal load reaches 130 nN, the lateral force increases sharply, until at very high loads (> 250 nN) the lateral force increases at a similar rate to neat hexadecane and the lower concentration IL/oil mixtures. These data indicate that while ions are adsorbed to the surface at this IL concentration, the adsorbed ion density is relatively low, resulting in a poorly formed boundary layer. Thus the breakpoint, the critical normal load at which the lateral force increases rapidly, is lower than neat $P_{6,6,6,14} (iC_8)_2PO_2$. At higher IL concentrations (≥ 1 mol%), the friction coefficients before breakpoints are similar to that of the neat $P_{6,6,6,14} (iC_8)_2PO_2$, *cf.* Table 1 and Figure 3. The breakpoint for 1 mol% $P_{6,6,6,14} (iC_8)_2PO_2$ /hexadecane mixture occurs at 170 nN, while for 5 mol% and neat $P_{6,6,6,14} (iC_8)_2PO_2$ there is a breakpoint at approximately 230 nN. Generally the nanofriction data for $P_{6,6,6,14} (iC_8)_2PO_2$ /hexadecane mixtures on alumina shows similar trends with those obtained on silica and titania.^{41, 44, 46} At low bulk IL concentrations (≤ 1.0 mol%) the lower surface excess of ions leads to a lower density of ions in the boundary layer which the tip displaces at high force. At higher IL concentrations, the higher surface excess leads to a more robust boundary layer, and the lateral forces are similar to that of the neat IL. Thus, in general, lateral forces decrease and the breakpoint increases with IL concentration. A clear exception to this trend is 2 mol% $P_{6,6,6,14} (iC_8)_2PO_2$, with lower lateral forces than all other $P_{6,6,6,14} (iC_8)_2PO_2$ /hexadecane concentrations and no obvious breakpoint. The reason for this will be discussed later in the manuscript.

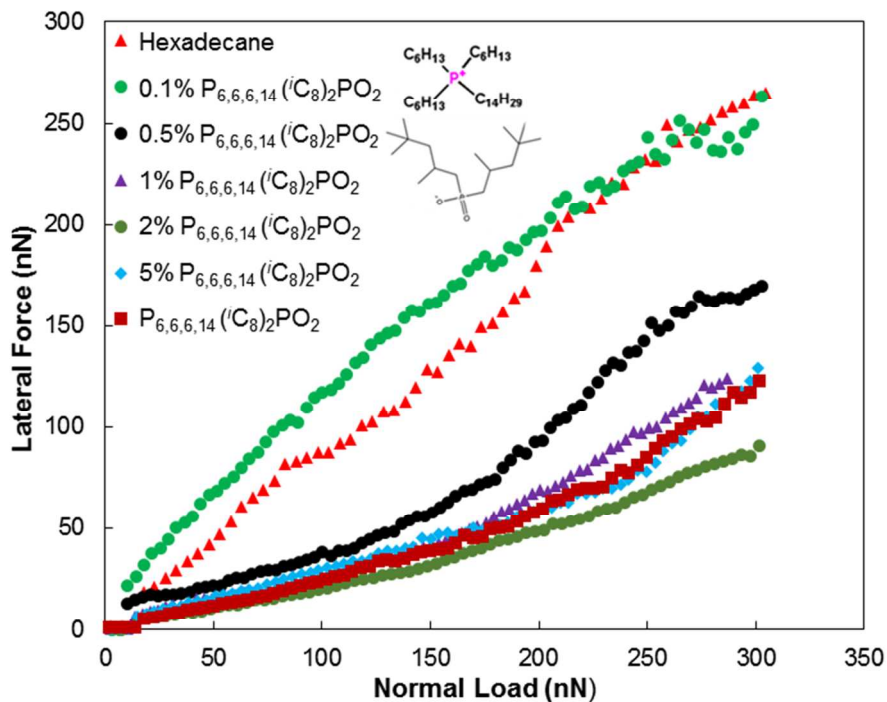


Figure 1. Lateral Force vs normal load in $P_{6,6,6,14}(iC_8)_2PO_2$ /hexadecane mixtures for an AFM tip ($r \approx 8$ nm) sliding on an alumina surface at $6 \mu m s^{-1}$. The structure of $P_{6,6,6,14}(iC_8)_2PO_2$ is shown in the inset.

Table 1. Friction coefficients (μ) for $P_{6,6,6,14}(iC_8)_2PO_2$ /hexadecane mixtures before and after breakpoints.

mol% $P_{6,6,6,14}(iC_8)_2PO_2$	Hexadecane	0.10%	0.50%	1%	2%	5%	$P_{6,6,6,14}(iC_8)_2PO_2$
μ (before break)	1.0 ± 0.3	0.95 ± 0.04	0.30 ± 0.02	0.28 ± 0.02	0.26 ± 0.02	0.28 ± 0.01	0.25 ± 0.01
μ (after break)			0.76 ± 0.09	0.61 ± 0.04		0.74 ± 0.09	0.55 ± 0.04
Breakpoint (nN)	n/a	n/a	130	170	n/a	230	230

$P_{6,6,6,14}$ DEHP

Friction measured in $P_{6,6,6,14}$ DEHP is very low ($\mu = 0.21 \pm 0.02$). At high loads (~ 230 nN), there is a sharp increase in the lateral force (Figure 2), similar to what is seen for $P_{6,6,6,14}(iC_8)_2PO_2$. At low to medium loads, an adsorbed IL boundary layer between the surfaces prevents direct

contact; however, at high loads, the AFM tip pushes through the boundary layer and it cannot reform within the time frame of the experiment, resulting in increased friction.

At low concentrations (< 0.1 mol%), the friction of $P_{6,6,6,14}$ DEHP/hexadecane mixtures is higher than that of neat $P_{6,6,6,14}$ DEHP but lower than that of hexadecane. In $P_{6,6,6,14}$ DEHP/hexadecane mixtures there are no obvious breakpoints up to 300 nN. Furthermore, when the concentration of $P_{6,6,6,14}$ DEHP is 0.1 mol% and higher, the lateral forces and friction coefficients are significantly reduced and comparable to those of neat $P_{6,6,6,14}$ DEHP before the breakpoint, *cf.* Figure 2 and Figure 3. These results indicate the formation of robust IL ion boundary layers at such IL concentrations. Interestingly, as observed for the $P_{6,6,6,14}$ (iC_8)₂PO₂/hexadecane mixtures, 2 mol% $P_{6,6,6,14}$ DEHP ($\mu = 0.09$) exhibits superior tribological performance over any other solution examined, even neat $P_{6,6,6,14}$ DEHP ($\mu = 0.21$). Additionally, we have found 95 mol% $P_{6,6,6,14}$ DEHP (Figure S4) has no observable breakpoints up to a normal load of 300 nN. These data indicate that the integration of hexadecane in the boundary layer plays a vital part in the improvement of lubricity. Although the IL and solvent are different, these results are in line with Seddon *et al*'s findings on molecular ordering of diluted ILs at the interface, which show that a small amount of either solvent or solute could increase the strength of boundary layers.⁴⁸

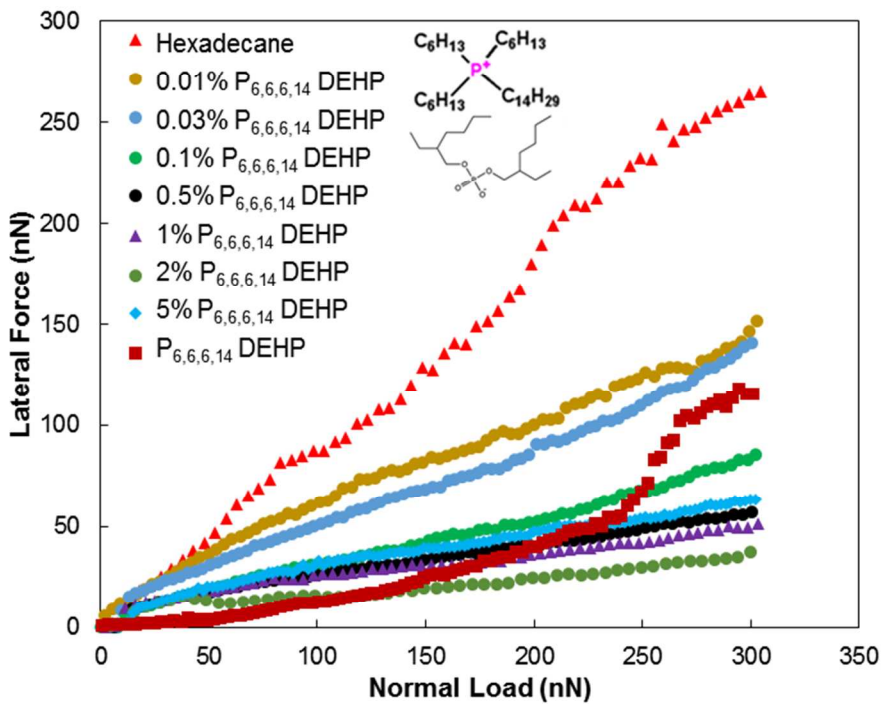


Figure 2. Lateral force vs normal load in P_{6,6,6,14} DEHP/hexadecane mixtures for an AFM tip ($r \approx 8$ nm) sliding on an alumina surface at $6 \mu\text{m s}^{-1}$. The structure of P_{6,6,6,14} DEHP is shown in the inset.

Table 2. Friction coefficients (μ) of P_{6,6,6,14} DEHP/Hexadecane mixtures on alumina.

mol% DEHP	0.01%	0.03%	0.10%	0.50%	1%	2%	5%	IL Breakpoint: 240 nN	
								before break point	after break point
μ	0.45	0.42	0.24	0.15	0.13	0.09	0.25	0.21	1.0
	± 0.02	± 0.02	± 0.02	± 0.01	± 0.01	± 0.01	± 0.02	± 0.02	± 0.1

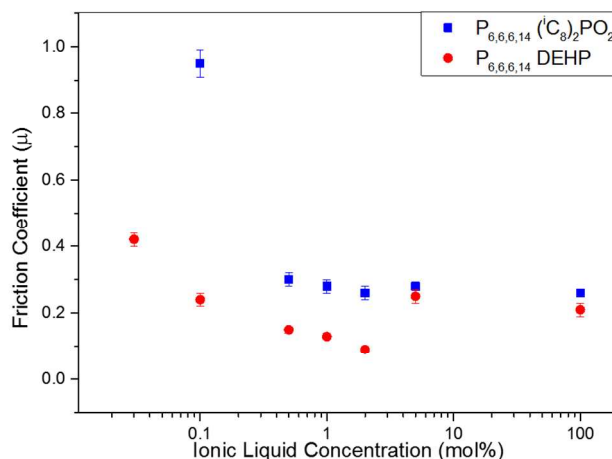


Figure 3. Friction coefficients of $P_{6,6,6,14} (iC_8)_2PO_2$ /hexadecane and $P_{6,6,6,14} DEHP$ /hexadecane mixtures on alumina before the break point.

Comparison of $P_{6,6,6,14} (iC_8)_2PO_2$ and $P_{6,6,6,14} DEHP$

ILs reduce friction by adsorbing to the solid surface and providing a low shear boundary layer which prevents contact between surfaces. At high loads, this boundary layer can be ruptured, resulting in direct contact between the surfaces, and a sharp increase in friction as seen in Figure 1 and Figure 2. The load at which the boundary layer is ruptured depends on: (i) the strength of the interaction between the IL ions in the boundary layer and the surface, (ii) the surface excess of IL at the interface, and (iii) the mobility of ions into the boundary layer. The molecular structure of the IL determines (i) the strength of the interaction between the IL ions in the boundary layer and the surface. Furthermore, we have shown in previous studies⁴⁶ as well as this study that (ii) the surface excess of IL at the interface depends on the bulk IL concentration. Until now, the importance of (iii) the mobility of the ions into the boundary layer is not clear.

Both ILs in this study are highly viscous (> 1000 mPa·s) at room temperature,⁴⁹ which means that at high loads when the tip displaces the boundary layer IL ions, the ability of the ions to replenish the boundary layer is limited. As hexadecane is significantly less viscous than the ILs

investigated (3 mPa·s vs ~1000 mPa·s), inclusion of hexadecane within the boundary layer may increase the mobility of ions into the boundary layer, and thus once disrupted, the boundary layer is able to replenish immediately, similar to measurements in aqueous salt solutions.⁵⁰⁻⁵¹ This is likely why neat P_{6,6,6,14} DEHP presents an obvious breakpoint at a normal load of 240 nN, whereas 0.1 mol% to 95 mol% P_{6,6,6,14} DEHP/hexadecane mixtures do not (Figure 2 and Figure S4). However, the presence of hexadecane in the boundary layer may also reduce the packing and ordering of ions, thus reducing the resistance of the boundary layer under high normal loads. 2 mol% IL/hexadecane mixtures show the best lubricity because the boundary layer at this concentration is robust enough and replenishes quickly under high normal loads.

In the neat form, P_{6,6,6,14} DEHP shows lubricity comparable to P_{6,6,6,14} (ⁱC₈)₂PO₂ on alumina. However, when mixed with hexadecane, P_{6,6,6,14} DEHP is more effective than P_{6,6,6,14} (ⁱC₈)₂PO₂; the critical concentration for P_{6,6,6,14} DEHP/hexadecane (0.1 mol%) is much lower than that of P_{6,6,6,14} (ⁱC₈)₂PO₂ (1.0 mol%), as well as the friction coefficients of the same concentration, *cf.* Figure 3. This suggests that when mixed with hexadecane, P_{6,6,6,14} DEHP adsorbs more effectively to a charged alumina surface than P_{6,6,6,14} (ⁱC₈)₂PO₂, thereby creating a smoother and more robust ion boundary layer, leading to reduced friction. ILs have been reported to be only partially dissociated, with some neutral ion-pairs or clusters.⁵² Ion diffusion rates for different structured ILs vary, but are generally proportional to the electrical conductivity. The conductivities of P_{6,6,6,14} DEHP and P_{6,6,6,14} (ⁱC₈)₂PO₂ are the same, both are 3.6 μS/cm.²⁷ This means the diffusion rates of the two ILs in the neat form are comparable, which is consistent with the similar friction coefficients found in the neat form. However, the conductivity of P_{6,6,6,14} DEHP/oil mixture is about two orders of magnitude higher than that of P_{6,6,6,14} (ⁱC₈)₂PO₂/oil mixture.²⁷ This suggests that at the same IL concentration, a larger portion of P_{6,6,6,14} DEHP

dissociates to single ions than $P_{6,6,6,14} (iC_8)_2PO_2$ when dissolved in apolar oils, leading to higher ion adsorption at the interface and more robust boundary layers and thus low friction.

The nanotribology results for $P_{6,6,6,14} (iC_8)_2PO_2$ and $P_{6,6,6,14}$ DEHP dissolved in apolar base oils on alumina surfaces obtained in this study are generally consistent with previous macroscale tribology studies on steel-aluminum surfaces.²⁷ Both studies show that $P_{6,6,6,14} (iC_8)_2PO_2$ /oil mixtures present obvious breakpoints at longer distances whereas $P_{6,6,6,14}$ DEHP/oil mixtures do not. These results indicate that the structure and composition of the boundary layers for neat ILs and IL/oil mixtures on alumina are influenced by the composition of ILs. ILs with higher dissociation rates tend to form a more robust boundary layer and thus show higher lubricity.

Conclusion

The nanotribology of alumina surfaces lubricated with $P_{6,6,6,14} (iC_8)_2PO_2$ and $P_{6,6,6,14}$ DEHP in hexadecane were investigated using AFM. For both $P_{6,6,6,14} (iC_8)_2PO_2$ and $P_{6,6,6,14}$ DEHP, the neat IL is a much more effective lubricant in comparison to neat hexadecane as seen in previous studies. For IL/oil mixtures, $P_{6,6,6,14}$ DEHP in hexadecane lubricates the alumina surface better than $P_{6,6,6,14} (iC_8)_2PO_2$. Hexadecane is observed to have a strong influence on the integrity of the boundary layer in combination with $P_{6,6,6,14}$ DEHP or $P_{6,6,6,14} (iC_8)_2PO_2$, as the presence of hexadecane assists the replenishment of boundary layers, but reduces the ordering of ions in the boundary layers. For both ILs investigated, the best tribological performance for each IL, is observed for 2 mol% IL in hexadecane, as the boundary layer is robust enough and replenishes quickly. The outcomes of this study help to reveal the effects of IL structure and concentration on lubricity on alumina surfaces, which assists the commercialization of IL lubricants for aluminum and alloys.

Supporting Information

- **Figure S1.** AFM image of Alumina Surface
- **Figure S2.** Repeated friction curves for 0.5 mol% $P_{6,6,6,14}$ (iC_8)₂PO₂/hexadecane and 0.5 mol% $P_{6,6,6,14}$ DEHP/hexadecane mixtures
- **Figure S3.** Typical normal force versus apparent separation profile for an AFM tip approaching and retracting from an alumina surface in pure hexadecane (a), pure $P_{6,6,6,14}$ (iC_8)₂PO₂ (b) and $P_{6,6,6,14}$ DEHP (c) and 95 mol% $P_{6,6,6,14}$ DEHP/hexadecane mixture (d)
- **Figure S4.** Lateral force vs normal load for 95% $P_{6,6,6,14}$ DEHP in Hexadecane.
- **Table S1.** Zero shear viscosity of $P_{6,6,6,14}$ (iC_8)₂PO₂/hexadecane mixtures at 20 °C measured by an AR-G2 rheometer (TA instruments)
- **Table S2.** Zero shear viscosity of $P_{6,6,6,14}$ DEHP /hexadecane mixtures at 20 °C measured by an AR-G2 rheometer (TA instruments)

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References

1. Stachowiak, G. W.; Batchelor, A. W., *Engineering Tribology*; Butterworth-Heinemann, 2013.
2. Neale, M. J., *The Tribology Handbook*; Butterworth-Heinemann, 1995.
3. Wan, Y.; Cao, L.; Xue, Q., Friction and Wear Characteristics of Zddp in the Sliding of Steel against Aluminum Alloy. *Tribol. Inter.* **1997**, *30*, 767-772.
4. Shimizu, Y.; Spikes, H. A., The Influence of Aluminium–Silicon Alloy on Zddp Tribofilm Formation on the Counter-Surface. *Tribol. Lett.* **2017**, *65*, 137.
5. Wasserscheid, P.; Welton, T., *Ionic Liquids in Synthesis*; Wiley Online Library, 2008; Vol. 1.
6. Liu, W.; Ye, C.; Gong, Q.; Wang, H.; Wang, P., Tribological Performance of Room-Temperature Ionic Liquids as Lubricant. *Tribol. Lett.* **2002**, *13*, 81-85.
7. Ye, C.; Liu, W.; Chen, Y.; Yu, L., Room-Temperature Ionic Liquids: A Novel Versatile Lubricant. *Chem. Commun.* **2001**, 2244-2245.
8. Han, Y.; Qiao, D.; Zhang, S.; Feng, D., Influence of Phosphate and Phosphonate Ionic Liquid Structures on Lubrication for Different Alloys (Mg, Al, Cu). *Tribol. Inter.* **2017**, *114*, 469-477.
9. Minami, I., Ionic Liquids in Tribology. *Molecules* **2009**, *14*, 2286-2305.
10. Somers, A.; Howlett, P.; MacFarlane, D.; Forsyth, M., A Review of Ionic Liquid Lubricants. *Lubricants* **2013**, *1*, 3.
11. Somers, A. E.; Biddulph, S. M.; Howlett, P. C.; Sun, J.; MacFarlane, D. R.; Forsyth, M., A Comparison of Phosphorus and Fluorine Containing II Lubricants for Steel on Aluminium. *Phys. Chem. Chem. Phys.* **2012**, *14*, 8224-8231.
12. Somers, A. E.; Howlett, P. C.; Sun, J.; MacFarlane, D. R.; Forsyth, M., Transition in Wear Performance for Ionic Liquid Lubricants under Increasing Load. *Tribol. Lett.* **2010**, *40*, 279-284.
13. Perkin, S.; Albrecht, T.; Klein, J., Layering and Shear Properties of an Ionic Liquid, 1-Ethyl-3-Methylimidazolium Ethylsulfate, Confined to Nano-Films between Mica Surfaces. *Phys. Chem. Chem. Phys.* **2010**, *12*, 1243-1247.
14. Sweeney, J.; Hausen, F.; Hayes, R.; Webber, G. B.; Endres, F.; Rutland, M. W.; Bennewitz, R.; Atkin, R., Control of Nanoscale Friction on Gold in an Ionic Liquid by a Potential-Dependent Ionic Lubricant Layer. *Phys. Rev. Lett.* **2012**, *109*, 155502.
15. Carstens, T.; Gustus, R.; Hoeffft, O.; Borisenko, N.; Endres, F.; Li, H.; Wood, R. J.; Page, A. J.; Atkin, R., Combined Stm, Afm, and Dft Study of the Highly Ordered Pyrolytic Graphite/1-Octyl-3-Methyl-Imidazolium Bis (Trifluoromethylsulfonyl) Imide Interface. *J. Phys. Chem. C* **2014**, *118*, 10833-10843.
16. Li, H.; Wood, R. J.; Rutland, M. W.; Atkin, R., An Ionic Liquid Lubricant Enables Superlubricity to Be "Switched on" in Situ Using an Electrical Potential. *Chem. Comm.* **2014**, *50*, 4368-4370.
17. Elbourne, A.; Sweeney, J.; Webber, G. B.; Wanless, E. J.; Warr, G. G.; Rutland, M. W.; Atkin, R., Adsorbed and near-Surface Structure of Ionic Liquids Determines Nanoscale Friction. *Chem. Commun.* **2013**, *49*, 6797-6799.
18. Greaves, T. L.; Weerawardena, A.; Krodziewska, I.; Drummond, C. J., Protic Ionic Liquids: Physicochemical Properties and Behavior as Amphiphile Self-Assembly Solvents. *J. Phys. Chem. B* **2008**, *112*, 896-905.
19. Triolo, A.; Russina, O.; Bleif, H. J.; DiCola, E., Nanoscale Segregation in Room Temperature Ionic Liquids. *J. Phys. Chem. B* **2007**, *111*, 4641-4644.
20. Rogers, R. D.; Seddon, K. R., Ionic Liquids--Solvents of the Future? *Science* **2003**, *302*, 792-793.
21. Li, H.; Rutland, M. W.; Atkin, R., Ionic Liquid Lubrication: Influence of Ion Structure, Surface Potential and Sliding Velocity. *Phys. Chem. Chem. Phys.* **2013**, *15*, 14616-14623.

22. Carstens, T.; Hayes, R.; Abedin, S. Z. E.; Corr, B.; Webber, G. B.; Borisenko, N.; Atkin, R.; Endres, F., In Situ Stm, Afm and Dts Study of the Interface 1-Hexyl-3-Methylimidazolium Tris (Pentafluoroethyl) Trifluorophosphate/Au (111). *Electrochim. Acta* **2012**, *82*, 48-59.
23. Li, H.; Wood, R. J.; Endres, F.; Atkin, R., Influence of Alkyl Chain Length and Anion Species on Ionic Liquid Structure at the Graphite Interface as a Function of Applied Potential. *J. Phys.: Condens. Matter* **2014**, *26*, 284115.
24. Elbourne, A.; Voitchovsky, K.; Warr, G. G.; Atkin, R., Ion Structure Controls Ionic Liquid near-Surface and Interfacial Nanostructure. *Chem. Sci.* **2015**, *6*, 527-536.
25. Yang, X.; Meng, Y.; Tian, Y., Effect of Imidazolium Ionic Liquid Additives on Lubrication Performance of Propylene Carbonate under Different Electrical Potentials. *Tribol. Lett.* **2014**, *56*, 161-169.
26. Yu, B.; Bansal, D. G.; Qu, J.; Sun, X.; Luo, H.; Dai, S.; Blau, P. J.; Bunting, B. G.; Mordukhovich, G.; Smolenski, D. J., Oil-Miscible and Non-Corrosive Phosphonium-Based Ionic Liquids as Candidate Lubricant Additives. *Wear* **2012**, *289*, 58-64.
27. Somers, A. E.; Khemchandani, B.; Howlett, P. C.; Sun, J.; MacFarlane, D. R.; Forsyth, M., Ionic Liquids as Antiwear Additives in Base Oils: Influence of Structure on Miscibility and Antiwear Performance for Steel on Aluminum. *ACS Appl. Mater. Inter.* **2013**, *5*, 11544-11553.
28. Barnhill, W. C.; Qu, J.; Luo, H.; Meyer, H. M.; Ma, C.; Chi, M.; Papke, B. L., Phosphonium-Organophosphate Ionic Liquids as Lubricant Additives: Effects of Cation Structure on Physicochemical and Tribological Characteristics. *ACS Appl. Mater. Inter.* **2014**, *6*, 22585-22593.
29. Qu, J.; Bansal, D. G.; Yu, B.; Howe, J. Y.; Luo, H.; Dai, S.; Li, H.; Blau, P. J.; Bunting, B. G.; Mordukhovich, G., Antiwear Performance and Mechanism of an Oil-Miscible Ionic Liquid as a Lubricant Additive. *ACS Appl. Mater. Inter.* **2012**, *4*, 997-1002.
30. Qu, J.; Luo, H.; Chi, M.; Ma, C.; Blau, P. J.; Dai, S.; Viola, M. B., Comparison of an Oil-Miscible Ionic Liquid and Zddp as a Lubricant Anti-Wear Additive. *Tribol. Inter.* **2014**, *71*, 88-97.
31. Fan, M.; Yang, D.; Wang, X.; Liu, W.; Fu, H., Doss- Based Qails: As Both Neat Lubricants and Lubricant Additives with Excellent Tribological Properties and Good Detergency. *Ind. Eng. Chem. Res.* **2014**, *53*, 17952-17960.
32. Zhou, Y.; Dyck, J.; Graham, T. W.; Luo, H.; Leonard, D. N.; Qu, J., Ionic Liquids Composed of Phosphonium Cations and Organophosphate, Carboxylate, and Sulfonate Anions as Lubricant Antiwear Additives. *Langmuir* **2014**, *30*, 13301-13311.
33. Cai, Z.; Meyer, H. M.; Ma, C.; Chi, M.; Luo, H.; Qu, J., Comparison of the Tribological Behavior of Steel-Steel and Si 3 N 4-Steel Contacts in Lubricants with Zddp or Ionic Liquid. *Wear* **2014**, *319*, 172-183.
34. Jiménez, A. E.; Bermúdez, M. D.; Carrión, F. J.; Martínez-Nicolás, G., Room Temperature Ionic Liquids as Lubricant Additives in Steel-Aluminium Contacts: Influence of Sliding Velocity, Normal Load and Temperature. *Wear* **2006**, *261*, 347-359.
35. Kumeria, T.; Santos, A.; Losic, D., Nanoporous Anodic Alumina Platforms: Engineered Surface Chemistry and Structure for Optical Sensing Applications. *Sensors* **2014**, *14*, 11878-11918.
36. Thompson, G., Porous Anodic Alumina: Fabrication, Characterization and Applications. *Thin Solid Films* **1997**, *297*, 192-201.
37. Liu, H.; Bhushan, B., Nanotribological Characterization of Molecularly Thick Lubricant Films for Applications to MEMS/NEMS by AFM. *Ultramicroscopy* **2003**, *97*, 321-340.
38. Czichos, H., *Tribology: A Systems Approach to the Science and Technology of Friction, Lubrication, and Wear*; Elsevier, 2009; Vol. 1.
39. Walsh, R. B.; Howard, S. C.; Nelson, A.; Skinner, W. M.; Liu, G.; Craig, V. S. J., Model Surfaces Produced by Atomic Layer Deposition. *Chem. Lett.* **2012**, *41*, 1247-1249.
40. Atkin, R.; Warr, G. G., Structure in Confined Room-Temperature Ionic Liquids. *J. Phys. Chem. C* **2007**, *111*, 5162-5168.

41. Li, H.; Somers, A. E.; Rutland, M. W.; Howlett, P. C.; Atkin, R., Combined Nano- and Macrotribology Studies of Titania Lubrication Using the Oil-Ionic Liquid Mixtures. *ACS Sus. Chem. Eng.* **2016**, *4*, 5005-5012.
42. Ralston, J.; Larson, I.; Rutland, M. W.; Feiler, A. A.; Kleijn, M., Atomic Force Microscopy and Direct Surface Force Measurements. *Pure Appl. Chem.* **2005**, *77*, 2149-2170.
43. Hadfield, M.; Wang, Y., *Tribology in Environmental Design* Professional Engineering Publishing Ltd., 2003.
44. Li, H.; Somers, A. E.; Howlett, P. C.; Rutland, M. W.; Forsyth, M.; Atkin, R., Addition of Low Concentrations of an Ionic Liquid to a Base Oil Reduces Friction over Multiple Length Scales: A Combined Nano-and Macrotribology Investigation. *Phys. Chem. Chem. Phys.* **2016**, *18*, 6541-6547.
45. Pit, R.; Hervet, H.; Leger, L., Direct Experimental Evidence of Slip in Hexadecane: Solid Interfaces. *Phys. Rev. Lett.* **2000**, *85*, 980.
46. Li, H.; Cooper, P. K.; Somers, A. E.; Rutland, M. W.; Howlett, P. C.; Forsyth, M.; Atkin, R., Ionic Liquid Adsorption and Nanotribology at the Silica–Oil Interface: Hundred-Fold Dilution in Oil Lubricates as Effectively as the Pure Ionic Liquid. *J. Phys. Chem. Lett.* **2014**, *5*, 4095-4099.
47. Israelachvili, J. N., Forces between Surfaces in Liquids. *Adv. Colloid Interf. Sci.* **1982**, *16*, 31-47.
48. Jitvisate, M.; Seddon, J. R. T., Near-Wall Molecular Ordering of Dilute Ionic Liquids. *J. Phys. Chem. C* **2017**, *121*, 18593-18597.
49. Battez, A. H.; Bartolomé, M.; Blanco, D.; Viesca, J.; Fernández-González, A.; González, R., Phosphonium Cation-Based Ionic Liquids as Neat Lubricants: Physicochemical and Tribological Performance. *Tribol. Inter.* **2016**, *95*, 118-131.
50. Raviv, U.; Klein, J., Fluidity of Bound Hydration Layers. *Science* **2002**, *297*, 1540.
51. Jacob, K.; Uri, R.; Susan, P.; Nir, K.; Liraz, C.; Suzanne, G., Fluidity of Water and of Hydrated Ions Confined between Solid Surfaces to Molecularly Thin Films. *J. Phys.: Condens. Matter* **2004**, *16*, S5437.
52. Lee, A. A.; Vella, D.; Perkin, S.; Goriely, A., Are Room-Temperature Ionic Liquids Dilute Electrolytes? *J. Phys. Chem. Lett.* **2015**, *6*, 159-163.

TOC Graphic

